

PHY 115L

Final Project Guidelines

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March 6, 2025

1 Overview

The final project is an opportunity for you to explore a physics problem of your own choosing and apply the techniques of computational physics to solve, illustrate, or in some other way illuminate it. It can be connected to techniques or topics we studied in class, or it can strike out in a completely new direction. Your problem should be connected to quantum mechanics or classical mechanics, but I will interpret this scope quite broadly. The most important thing is that your topic is well chosen as something of interest to you, on which you can make reasonable progress.

In terms of size, this should be something like 50% of the scope of a typical homework. (This is because the homework lay out a very clear path, so progress is much more rapid.)

2 Examples

You are not required to take one of these examples, but here are some examples of reasonably scoped projects:

- Adding spin-orbit coupling to our numerical model for the neutron.
- Animating the time evolution of a wave packet in a quantum harmonic oscillator.
- Animating scattering of a wave packet.
- Plotting the 3-D solutions to the hydrogen atom in an illustrative manner.
- Exploring the “shooting method” to finding energy eigenstates.
- Develop a tool for calculating (exactly) the Clebsch-Gordan coefficients.